

Performance Analysis of Logistic Regression under Controlled Class Separation and Overlap

Saroar Jahan Shuba
Department of Mathematics
Lamar University
Texas, United States
sshuba@lamar.edu

Abstract—This project explores the behavior of logistic regression in a controlled three-dimensional setting using synthetically generated data. Two types of scenarios are considered: one where the classes are clearly separated and another where the classes overlap. In the non-overlapping case, the model achieves perfect classification on both training and testing data, showing that a simple linear boundary is sufficient when the data is well structured.

To better understand how separation affects performance, the distance between the class centers is varied systematically. The results show that as the distance increases, the misclassification error decreases rapidly and eventually approaches zero, confirming that class separation plays a key role in classification accuracy.

In the overlapping case, the data distributions introduce ambiguity, making perfect classification impossible. To handle this, the effect of regularization is examined by comparing a weak prior ($C = 10$) and a strong prior ($C = 0.01$). The results show that stronger regularization slightly increases training error but improves testing performance, highlighting the trade-off between model flexibility and generalization.

Overall, the study demonstrates that the performance of logistic regression depends heavily on the underlying data geometry, and that appropriate regularization can significantly improve model reliability when the data is not perfectly separable.

Index Terms—Logistic Regression, Classification, Class Separation, Overlapping Data, Misclassification Error, Regularization, Bias-Variance Tradeoff

I. INTRODUCTION

Logistic regression is one of the most widely used classification techniques in statistical learning due to its simplicity, interpretability, and solid theoretical foundation [4]. It models the probability of class membership by applying a sigmoid transformation to a linear combination of input features. Despite being a relatively simple linear model, it often provides strong performance and serves as an essential baseline for understanding classification problems [2].

In this study, we analyze the behavior of logistic regression within a controlled three-dimensional feature space. Synthetic datasets are generated to form two distinct classes, allowing us to directly examine how the geometric structure of the data influences classification performance. The primary goal is to evaluate misclassification error on both training and testing datasets and understand how well the model generalizes under different conditions.

Two experimental scenarios are considered. In the first scenario, the two classes are well separated, resulting in a linearly separable dataset. By varying the distance between class centers over a specified range, we investigate how increasing separation improves classification accuracy and reduces error. This provides a clear understanding of model behavior under ideal conditions where a linear decision boundary is sufficient.

In the second scenario, overlap between the two classes is introduced to reflect more realistic data conditions. In particular, the class center distance is fixed at $d = 4$ while the spread is increased to $\sigma = 1.8$, leading to significant overlap between the data distributions. In such cases, perfect classification is not achievable, and the model must balance errors across ambiguous regions of the feature space.

To address the risk of overfitting in the overlapping setting, regularization is incorporated through a Bayesian perspective. Specifically, an L_2 penalty is applied to the model parameters, which is equivalent to assuming a Gaussian prior on the coefficients [2]. This helps control model complexity and stabilizes the learned decision boundary.

Throughout the analysis, both training and testing misclassification errors are computed to assess model performance. This allows us to observe how data geometry and regularization jointly influence the bias-variance trade-off and overall classification effectiveness [2].

II. DATA GENERATION

To study the behavior of logistic regression in a controlled environment, we generate a synthetic dataset in a three-dimensional feature space. The dataset consists of two classes, each drawn from a multivariate Gaussian distribution.

We set the number of samples per class to $n = 500$, resulting in a total of $2n = 1000$ observations. The feature vectors for the two classes are generated as:

$$X_0 \sim \mathcal{N}(\mu_0, I), \quad X_1 \sim \mathcal{N}(\mu_1, I)$$

where I denotes the 3×3 identity covariance matrix. This ensures that all features are independent and share the same variance across dimensions.

A. Non-Overlapping Configuration

In the first scenario, the two classes are well separated in the feature space. The class centers are defined as:

$$\mu_0 = (0, 0, 0), \quad \mu_1 = (5, 5, 5)$$

This configuration produces two clearly distinct clusters with negligible overlap, resulting in a linearly separable dataset. Under such conditions, logistic regression is expected to perform extremely well, as a linear decision boundary is sufficient to separate the classes.

To further analyze the effect of class separation, we perform a distance-based experiment by varying the position of the second class center. Specifically, we define:

$$\mu_1 = (d, d, d)$$

and allow d to vary over the range:

$$d \in \{0.5, 1.0, 1.5, \dots, 6.0\}$$

By increasing d , the two class distributions gradually move farther apart. This transition enables us to observe how classification performance improves as the degree of separation increases, and how misclassification error decreases as the data becomes more distinguishable.

B. Overlapping Configuration

To create a more realistic and challenging classification setting, we construct a second dataset where the two classes are not perfectly separable. This is achieved by increasing the spread of each class while controlling the distance between their centers.

In this configuration, we use:

- Spread parameter $\sigma = 1.8$, which determines how widely the data points are dispersed
- Center distance $d = 4$, which controls how far apart the two class centers are

The feature vectors are generated as:

$$X_0 = \sigma Z_0 + \mu_0, \quad X_1 = \sigma Z_1 + \mu_1$$

where $Z_0, Z_1 \sim \mathcal{N}(0, I)$, and the class means are defined as:

$$\mu_0 = (0, 0, 0), \quad \mu_1 = (4, 4, 4)$$

With this setup, the two distributions partially overlap in the feature space. The relative relationship between the center distance and spread, expressed as $d/\sigma = 4/1.8 \approx 2.22$, determines how much the classes intersect. Compared to the non-overlapping case, this configuration introduces ambiguity near the boundary region, making perfect classification impossible.

C. Dataset Construction

The complete dataset is formed by combining the samples from both classes:

$$X = \begin{pmatrix} X_0 \\ X_1 \end{pmatrix} \in \mathbb{R}^{2n \times 3}, \quad y = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{R}^{2n}$$

To evaluate model performance, the dataset is split into training and testing subsets using a 70/30 hold-out strategy.

A fixed random seed of 42 is used to ensure reproducibility, resulting in 700 training samples and 300 testing samples. This split allows us to assess both how well the model fits the data and how effectively it generalizes to unseen observations.

III. METHODOLOGY

A. Logistic Regression Model

We model the probability of class membership using logistic regression, a widely used probabilistic classification approach [4]. For a given input vector X , the probability of belonging to class 1 is defined as:

$$P(Y = 1 | X) = \sigma(\beta_0 + \beta^T X)$$

where $\sigma(z)$ denotes the sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

This transformation maps the linear combination of features into a probability value between 0 and 1.

The model parameters, consisting of the coefficient vector $\beta = (\beta_1, \beta_2, \beta_3)$ and the intercept β_0 , are estimated by maximizing the likelihood of the observed data [4]. In practice, this is equivalent to minimizing the negative log-likelihood (also known as cross-entropy loss), given by:

$$\mathcal{L}_{\text{MLE}}(\beta) = - \sum_{i=1}^{2n} [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

where $p_i = P(Y = 1 | X_i)$ represents the predicted probability for the i -th observation.

In this study, model fitting is performed using standard optimization routines that iteratively adjust the parameters to minimize the loss function. This approach ensures that the estimated model provides the best possible fit to the training data under the logistic regression framework [2].

B. Bayesian Prior and Regularization

In situations where the class distributions overlap, standard maximum likelihood estimation (MLE) can lead to overfitting, as the model may attempt to capture noise present in the training data. On the other hand, in perfectly separable cases, the estimated coefficients can grow very large in magnitude, since the optimization process pushes predicted probabilities toward extreme values of 0 and 1 [2].

To address these issues, we adopt a Bayesian perspective by introducing a prior distribution on the model parameters. Specifically, we assume a zero-mean isotropic Gaussian prior on the coefficient vector:

$$\beta \sim \mathcal{N}(0, \tau^2 I)$$

This formulation shifts the estimation procedure from maximum likelihood to maximum a posteriori (MAP) estimation. The resulting objective function becomes:

$$\mathcal{L}_{\text{MAP}}(\beta) = - \sum_{i=1}^{2n} [y_i \log p_i + (1 - y_i) \log(1 - p_i)] + \frac{\lambda}{2} \|\beta\|^2$$

where the regularization parameter λ is inversely related to the prior variance ($\lambda \propto 1/\tau^2$). This additional penalty term discourages large coefficient values and helps stabilize the model [2]. From an optimization perspective, this is equivalent to L_2 regularization, which is commonly used to control model complexity [1].

In practice, regularization strength is controlled through the parameter $C = 1/\lambda$. A smaller value of C corresponds to stronger regularization, resulting in greater shrinkage of the coefficients, while a larger value allows more flexibility in fitting the data. The intercept term β_0 is not penalized.

In this study, we consider two different regularization settings:

- **Weak regularization:** $C = 10$, allowing the model to fit the data more closely
- **Strong regularization:** $C = 0.01$, enforcing stronger shrinkage of the coefficients

C. Decision Boundary

The decision boundary of the logistic regression model is defined as the set of points where the predicted probability is equal to 0.5. This occurs when the linear predictor equals zero [4]:

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 = 0$$

Rearranging the equation, we can express the boundary explicitly as:

$$x_3 = -\frac{\beta_0 + \beta_1 x_1 + \beta_2 x_2}{\beta_3}$$

This equation represents a plane in three-dimensional space that separates the two classes. The orientation and position of this plane depend on the learned parameters and are directly influenced by both the data distribution and the regularization strength.

D. Prediction and Model Evaluation

Predicted class labels are obtained by applying a threshold to the estimated probabilities from the logistic regression model. Specifically, an observation is assigned to class 1 if the predicted probability is at least 0.5, and to class 0 otherwise [4]:

$$\hat{y}_i = \begin{cases} 1 & \text{if } \hat{p}_i \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

where $\hat{p}_i = P(Y = 1 | X_i)$ is computed using the sigmoid function.

To evaluate model performance, we compute classification accuracy, defined as the proportion of correctly classified observations:

$$\text{Accuracy} = \frac{1}{2n} \sum_{i=1}^{2n} \mathbf{1}(\hat{y}_i = y_i)$$

where $\mathbf{1}(\cdot)$ is the indicator function. The misclassification error is then given by:

$$\text{Error} = 1 - \text{Accuracy}$$

Both training and testing errors are reported to assess how well the model generalizes to unseen data. In the overlapping scenario, any noticeable gap between training and testing performance provides evidence of overfitting, while similar values indicate that the model is able to generalize effectively despite the presence of class ambiguity [2].

E. Visualization

To better understand the classification behavior, the data points and corresponding decision boundaries are visualized in three-dimensional space. Each class is represented as a scatter of points, while the decision boundary is illustrated as a plane determined by the learned model parameters.

IV. RESULTS AND DISCUSSION

A. Case 1: Non-Overlapping Data

In the first scenario, the class centers are located at $(0, 0, 0)$ and $(5, 5, 5)$, producing a clearly separable dataset. Under this configuration, the logistic regression model achieves perfect classification on both the training and testing sets.

The estimated model parameters are:

$$\beta = [1.0824, 1.0459, 0.8775], \quad \beta_0 = -7.4768$$

The coefficients are fairly balanced across the three features, indicating that each dimension contributes similarly to the classification decision. This is consistent with the symmetric structure of the data. The intercept term shifts the decision boundary so that it lies approximately between the two class centers.

Both training and testing accuracies are equal to 1.0, corresponding to zero misclassification error. This confirms that the dataset is perfectly linearly separable, and the model is able to identify a decision boundary that completely distinguishes between the two classes.

This result is further supported by the visualization in Fig. 1, where the separating plane lies in the empty region between the two clusters. The absence of overlap allows the model to achieve optimal performance without any ambiguity.

Logistic Regression (Non-Overlapping Data)



Fig. 1. Non-overlapping class structure in three-dimensional space. Due to strong separation between the clusters, the model identifies a linear decision boundary that perfectly partitions the data without any classification errors.

B. Effect of Class Separation on Misclassification Error

To better understand how class separation influences model performance, we conducted a distance-based experiment by varying the center distance d from 0.5 to 6.0. The results,

presented in Fig. 2, show a clear decreasing trend in both training and testing misclassification error as d increases.

When the distance is small (e.g., $d < 1.5$), the two classes overlap significantly, leading to higher error rates. In this region, the model struggles to identify a reliable separating boundary due to the ambiguity between the classes. As d increases, the overlap gradually decreases, making it easier for the model to distinguish between the two groups.

Around $d \approx 3.0$ and beyond, both training and testing errors approach zero, indicating that the dataset has effectively become linearly separable. In this regime, logistic regression is able to learn a decision boundary that cleanly separates the classes.

Additionally, the testing error closely follows the training error across all values of d , with only minor fluctuations due to sampling variability. As the separation increases, the gap between training and testing performance becomes negligible, reflecting improved generalization and reduced inherent ambiguity in the data [2].



Fig. 2. Training and testing misclassification error as a function of the class center distance d . As d increases from 0.5 to 6.0, both errors decrease steadily and approach zero once the classes become well separated (approximately $d \geq 3$), illustrating how greater separation improves classification performance.

C. Overlapping Data and Regularization Effects

Unlike the separable case, perfect classification is theoretically unattainable here due to overlapping class distributions. Table I and Table II summarize performance metrics for all experimental settings.

For the overlapping scenario ($\sigma = 1.8$, $d = 4$), we examine how regularization influences model performance.

1) *Irreducible Error (Bayes Error)*: In the overlapping setting, even an optimal classifier cannot achieve zero misclassification error due to the inherent overlap between the class-conditional distributions. This unavoidable error is known as the *Bayes error*, which represents the minimum achievable error determined by the data-generating process [2].

Consequently, the non-zero testing error observed in this case is not solely due to limitations of the logistic regression model, but is fundamentally imposed by the structure of the data itself. The model therefore learns a decision boundary that balances classification errors across both classes rather than eliminating them entirely.

2) *Model Behavior in the Overlapping Setting*: The estimated parameters for the overlapping dataset are:

$$\beta = [1.4969, 1.5142, 0.8381], \quad \beta_0 = -5.8274$$

TABLE I
TRAINING AND TESTING ACCURACY ACROSS CASES

Case	Training Acc.	Testing Acc.
Non-Overlapping ($C = 1.0$)	1.0000	1.0000
Overlapping ($C = 1.0$)	0.9657	0.9600

TABLE II
MISCLASSIFICATION ERROR RATES ACROSS CASES

Case	Training Error	Testing Error
Non-Overlapping ($C = 1.0$)	0.0000	0.0000
Overlapping ($C = 1.0$)	0.0343	0.0400

Compared to the non-overlapping case, the coefficients remain relatively large but are more balanced across features, reflecting the increased difficulty of separating the two classes. Due to the overlap in the data, the model cannot form a perfectly clean boundary and instead learns a compromise that minimizes overall classification error.

The training error is 0.0343, while the testing error is slightly higher at 0.0400. This small difference indicates a mild generalization gap, which is expected when the data contains inherent ambiguity. Unlike the separable case, increasing the steepness of the decision boundary cannot eliminate misclassification entirely, as some points from different classes occupy similar regions in the feature space [2].

The corresponding 3D decision boundary is illustrated in Fig. 3, where the separating plane passes through regions containing points from both classes, highlighting the unavoidable classification uncertainty in this setting.

Logistic Regression (Overlapping Data)



Fig. 3. Visualization of the overlapping dataset in $\mathbb{R}^3$. The decision boundary passes through regions where the two classes overlap, reflecting the inherent ambiguity in the data. Training error = 0.0343 and testing error = 0.0400, indicating a small generalization gap.

3) *Regularization Effect in the Overlapping Setting*: In the overlapping scenario, the logistic regression model is trained using a fixed regularization parameter ($C = 1.0$), which provides a balance between model flexibility and stability. The estimated parameters are:

$$\beta = [1.4969, 1.5142, 0.8381], \quad \beta_0 = -5.8274$$

The coefficient magnitudes reflect moderate regularization, allowing the model to capture the overall structure of the data without overfitting to noise. Unlike stronger regularization

settings, the coefficients are not heavily shrunk, but they are also not excessively large, indicating a balanced fit.

The training error is 0.0343 and the testing error is 0.0400. The slightly higher testing error indicates a small generalization gap, which is expected due to the inherent overlap between the classes. This behavior highlights the role of regularization in maintaining stability while dealing with ambiguous data distributions [2].

4) *Effect of Regularization Strength:* To examine how regularization influences model performance, we vary the parameter $C = 1/\lambda$ over the set $\{0.001, 0.01, 0.05, 0.1, 0.5, 1, 5, 10, 50, 100\}$. The results are presented in Fig. 4.

For very small values of C (e.g., 0.001 and 0.01), corresponding to strong regularization, both training and testing errors are relatively higher, indicating that the model is too constrained to capture the underlying structure of the data effectively. As C increases, the training error decreases steadily, reflecting increased model flexibility.

The testing error initially decreases as C increases, reaching its minimum around moderate values of C (approximately between 0.05 and 0.1). Beyond this range, the testing error increases slightly and then stabilizes, suggesting that further increases in model flexibility do not lead to meaningful improvements in generalization.

Overall, the results indicate that moderate regularization provides the best balance between fitting the training data and maintaining strong generalization performance.

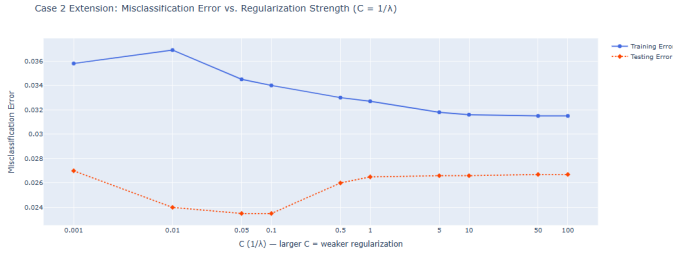


Fig. 4. Misclassification error as a function of regularization strength $C = 1/\lambda$ for the overlapping dataset. As C increases, the training error decreases steadily, while the testing error initially decreases, reaches a minimum at moderate values of C , and then stabilizes.

D. Parameter Estimation and Decision Boundaries

The 3D visualizations shown in Figs. 1 and 3 illustrate how the logistic regression decision boundary behaves under different data configurations. In the non-overlapping case, the decision plane clearly separates the two clusters by passing through the empty region between them, resulting in perfect classification.

In the overlapping setting, the decision boundary passes through regions where the two classes intersect, reflecting the inherent ambiguity in the data. As a result, some misclassification is unavoidable, and the model learns a boundary that balances errors across both classes rather than perfectly separating them.

The orientation of the boundary is determined by the estimated coefficients, while its position is influenced by the intercept term. In the overlapping case, the boundary cuts through denser regions of the data, highlighting the difficulty of achieving perfect separation when the class distributions overlap [2].

Although separate 3D visualizations are not shown for both regularization settings, the numerical results indicate that stronger regularization produces smaller coefficient magnitudes and a more conservative decision boundary, whereas weaker regularization allows a sharper and more flexible separating surface.

E. Overall Observations

The empirical results highlight several key observations. First, logistic regression achieves perfect classification when the class separation is sufficiently large, as demonstrated in the non-overlapping case where both training and testing errors are zero. As the distance between class centers decreases, the misclassification error increases due to greater overlap between the distributions, with minor fluctuations arising from sampling variability [4].

Second, in the overlapping setting, perfect classification is no longer possible due to the inherent ambiguity in the data. This leads to non-zero training and testing errors, even when the model is well-fitted. The relatively small gap between training and testing performance indicates that the model generalizes well despite the presence of overlap [2].

Third, the effect of regularization is clearly illustrated in both the regularization sweep (Fig. 4) and the summary comparison (Fig. 5). Stronger regularization reduces coefficient magnitudes, leading to a slight increase in training error but an improvement in testing performance. In contrast, weaker regularization allows the model to fit the training data more closely but results in slightly higher testing error. This behavior reflects the bias-variance trade-off and highlights the importance of selecting an appropriate value of C [2].

Overall, these results demonstrate the combined influence of data geometry and regularization on the effectiveness of logistic regression models [2].



Fig. 5. Comparison of training and testing misclassification errors for the non-overlapping and overlapping datasets. The non-overlapping case achieves zero error, while the overlapping case exhibits non-zero error due to inherent class overlap. The close agreement between training and testing errors in the overlapping setting indicates good generalization performance.

V. CONCLUSION

This project examined the behavior of logistic regression in a controlled three-dimensional feature space using simulated datasets with varying levels of class separation. By analyzing both non-overlapping and overlapping configurations, we observed how the structure of the data directly influences classification performance.

In the non-overlapping setting, logistic regression achieves perfect performance, with zero misclassification error on both training and testing data. The model successfully identifies a linear decision boundary that cleanly separates the two classes, demonstrating its effectiveness when the data is well-separated.

As the separation between class centers decreases, the classification task becomes more challenging. In overlapping scenarios, ambiguity in the feature space makes perfect separation impossible. Consequently, both training and testing errors increase, reflecting the inherent uncertainty in the data.

The results further demonstrate that regularization plays an important role in controlling model behavior. Stronger regularization reduces the magnitude of the coefficients, leading to a slight increase in training error but improved testing performance, as observed in the comparison between weak and strong regularization settings. The regularization sweep also shows that moderate values of the parameter C provide the best balance between model flexibility and generalization performance.

Overall, this study highlights that while logistic regression is a simple and effective classification method, its success depends strongly on the underlying geometry of the data. Class separation, overlap, and appropriate regularization all play a critical role in determining model performance in practical applications.

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